

Effects of Sex and Cuff Pressure on Physiological Responses during Blood Flow Restriction Resistance Exercise in Young Adults

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¹Integrated Laboratory of Exercise and Applied Physiology (iLEAP), Department of Health, Kinesiology and Sport, College of Education and Professional Studies, University of South Alabama, Mobile, AL; ²College of Medicine, University of South Alabama, Mobile, AL; ³Division of Kinesiology, School of Kinesiology and Physical Therapy, University of Central Florida, Orlando, FL; and ⁴Florida Space Institute, University of Central Florida, Orlando, FL

ABSTRACT

GRAY, S. M., A. M. CUOMO, C. E. PROPPE, M. K. TRAYLOR, E. C. HILL, and J. L. KELLER. Effects of Sex and Cuff Pressure on Physiological Responses during Blood Flow Restriction Resistance Exercise in Young Adults. *Med. Sci. Sports Exerc.*, Vol. 55, No. 5, pp. 920–931, 2023. **Purpose:** The purpose of this study was to examine the physiological responses resulting from an acute blood flow restriction resistance exercise bout with two different cuff pressures in young, healthy men and women. **Methods:** Thirty adults (18–30 yr) completed a bilateral leg extension blood flow restriction bout consisting of four sets (30–15–15–15 repetitions), with cuffs applied at pressures corresponding to 40% and 60% of the minimum arterial occlusion pressure (AOP) needed to completely collapse the femoral arteries. During each of these conditions (40% and 60% AOP), physiological measures of near-infrared spectroscopy (NIRS) and EMG amplitude (EMG AMP) were collected from the dominant or nondominant vastus lateralis. After each set, ratings of perceived exertion (RPE) were collected, whereas only at baseline and at the end of the bout, mean arterial pressure (MAP) was assessed. Separate mixed-factorial ANOVA models were used to examine mean differences in the change in EMG AMP and NIRS parameters during each set. The absolute RPE and MAP values were also examined with separate ANOVAs. A P value ≤ 0.05 was considered statistically significant. **Results:** Regardless of sex or cuff pressure, the change in EMG AMP was lower in set 1 (14.8%) compared with the remaining sets (22.6%–27.0%). The 40% AOP condition elicited the greatest changes in oxy[heme] and deoxy[heme], while also providing lower RPEs. For MAP, there was an effect for time such that MAP increased from preexercise (87.5 ± 4.3 mm Hg) to postexercise (104.5 ± 4.1 mm Hg). **Conclusions:** The major findings suggested that the 40% AOP condition permitted the greatest amount of recovery during the interset rest. In addition, there did not seem to be any meaningful sex-related difference in this sample of young healthy adults. **Key Words:** SEX DIFFERENCES, NEAR-INFRARED SPECTROSCOPY, EXERCISE PRESSOR REFLEX, ELECTROMYOGRAPHY, BILATERAL EXERCISE, RATINGS OF PERCEIVED EXERTION

Investigations of blood flow restriction (BFR) combined with resistance exercise continue to garner substantial interest within athletic, clinical, and recreational populations (1–6). Commonly, BFR is the practice of applying an inflated, pneumatic cuff to the most proximal portion of a limb to, in theory, completely occlude venous outflow while only partially reducing arterial inflow (1,3). It is interesting to note, however, that despite its growing popularity, no consensus-related cuff pressure has been reached (3,4,6,7). Most recently, Das and

Paton (4) suggested that training-induced adaptations likely only occur when using a range between 50% and 80% of the minimum arterial occlusion pressure (AOP) necessary to completely prevent inflow (4). Hypothetically, below this range may be insufficient to induce metabolic stress required for adaptation, whereas pressures above this recommendation may promote premature fatigue (4,6) before sufficient physiological stress. Although this is interesting work that provided testable hypotheses (4), it has previously been shown that cuff pressures corresponding to 40% AOP were capable of producing similar adaptations in muscular strength and indices of hypertrophy compared with conditions using an equal training load but higher cuff pressures (8–10). However, it should be noted that evidence exists showing that higher cuff pressures (e.g., 80% AOP) may be necessary to improve important cardiovascular characteristics such as vascular tone (11). Based on this collection of reports, it seems that a universally agreed-upon pressure has still not been identified, and thus, studies are still needed to contribute to the understanding of what precise cuff pressure or range of pressures

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is most efficacious, especially across a wide range of physiological responses. Perhaps, of even greater priority than determining the most ideal cuff pressure, there is currently a severe underrepresentation of women in biomedical research, including BFR investigations (12). Counts and coauthors (12) stated that only 29% of acute BFR research participants were women, which is concerning because of the little that is currently known regarding how the BFR stimulus influences female-related exercise responses, whereas there are known sex differences in muscle morphology, function, and plasticity including differences in gene expression after acute resistance exercise (13,14). Unfortunately, despite the repeated calls to increase the number of women in related studies, there is no evidence to suggest that parity has been reached between men and women. Therefore, until more data showing acute, physiological responses in a cuff- and sex-specific manner exist, appropriate applications of BFR will remain limited.

The importance of studying the effects of cuff pressure and sex in relation to BFR exercise includes achieving a better understanding, and perhaps control, of the wide range of possible physiological and perceptual responses. Recently, it was stated that there is a critical need to report acute physiological responses resulting from exercises performed with different cuff pressures (6). Electromyography (EMG) has often been used in investigations to examine various acute neuromuscular responses such as skeletal muscle excitability (1,15–18). For example, as a result of 75 leg extensions distributed across four sets (1×30 , 3×15), we reported that the muscle excitation (EMG amplitude (EMG AMP)) recorded during a brief maximum isometric task decreased pretest to posttest after the BFR exercise bout (60% AOP), but not after the non-BFR condition (1). This was interpreted as BFR-inducing an attenuation of local muscle excitation during the brief, maximum effort task likely provoked by metabolic stress including muscle acidosis and the consequential feedback from group III/IV muscle afferents (19). During submaximal efforts, however, Yasuda et al. (20,21) have stated that restricting blood flow via external compression increases integrated EMG mediated by the central nervous system in response to group III/IV feedback and perhaps because of a greater reliance on oxygen-independent metabolic pathways compared with free-flow conditions (22). To advance these findings, near-infrared spectroscopy (NIRS) can be collected alongside EMG to provide additional context into the changes of the skeletal muscle environment resulting from various exercise modalities (7,17,23). For example, with a sample of 11 young, healthy men, it was demonstrated that a single set of 30 unilateral, isokinetic leg extensions with occlusion pressures ranging from 0% to 100% AOP caused tissue oxygenation to decrease as cuff pressures increased (17). In addition, these investigators reported that the rate of increase (i.e., slope) in EMG AMP was greatest during 100% AOP, but that there were no differences in this rate between the 60% and 80% AOP conditions (17). Although these are important findings, it is currently unclear if these results extend to women, additional sets such as during common BFR protocols, and/or more popular exercise modalities

(e.g., bilateral and dynamic constant external resistance (DCER) muscle actions).

Additional parameters that warrant continued investigation include mean arterial pressure (MAP) responses and self-reported ratings of perceived exertion (RPE). It is possible that using compression devices may induce unfavorable increases in blood pressure (BP) and subsequently cause adverse vascular events, especially in individuals with signs and symptoms of cardiovascular disease (5,24). Of greatest relevance, concerns have been presented related to an augmented exercise pressor reflex (EPR) after BFR exercise because of a greater accumulation of metabolic by-products (2,19,24). It is curious, however, that women are known to have an attenuated EPR compared with men (25,26), yet no known studies have evaluated this under BFR exercise condition, especially when considering the typical BFR protocol consisting of 75 repetitions. Furthermore, in addition to possible negative health outcomes (e.g., unfavorable increases in BP), higher cuff pressures have been shown to provoke greater RPE and discomfort (27,28), which presumably reduce overall enjoyment and future adherence (29). However, there is still a gap in the associated literature concerning if and to what extent cuff pressure interacts with sex to provoke specific responses in MAP or RPE during BFR exercise. Determining this would further advance the ability of exercise professionals and clinicians to prescribe BFR exercise bouts.

As presented previously, there are limited existing data related to sex-specific physiological responses to BFR resistance exercise with standardized, relative occlusion pressures, especially during bilateral tasks, which remain quite popular in nonresearch settings compared with unilateral exercise (30). Therefore, the purpose of the current investigation was to evaluate the physiological responses resulting from an acute bilateral, BFR resistance exercise bout with two commonly reported relative cuff pressures in healthy, men and women. Based on previous reports (1,7,15,17,20,21,25), it was hypothesized that the higher cuff pressure would elicit the greatest change in EMG AMP as well as NIRS-derived parameters such as oxygenated hemoglobin (oxy[heme]) and deoxygenated hemoglobin (deoxy[heme]). It was also hypothesized that the men would exhibit the greatest change in MAP, NIRS parameters, and EMG AMP, whereas the women would report the greatest RPE (31).

METHODS

Ethics Approval

This study was performed according to the ethics standards established by the Declaration of Helsinki 2013 and was approved by the local Institutional Review Board for Human Subjects at the University of South Alabama (IRB no. 20-089/1897727-2). The study was not registered in a database, and all participants gave signed, written informed consent before completing any experimental protocols. No data related to this project have previously been published or submitted in any form.

Experimental Design

To address the purpose of the current study, a randomized, crossover (40% and 60% AOP), between group (sex) experimental design was used. Healthy adults visited the Integrative Laboratory of Exercise and Applied Physiology (iLEAP) for a total of three visits: one familiarization and two testing. The testing visits were counterbalanced. During the familiarization visit, the AOP for each leg and one-repetition maximum (1RM) for the bilateral leg extension exercise were determined. The testing visits included the participants completing four sets of bilateral, leg extensions at a low load (30% 1RM) with either 40% or 60% AOP independently applied to each leg. In addition, physiological measurements (NIRS and EMG) were collected from either the dominant or nondominant vastus lateralis. That is, at random, either the dominant or nondominant vastus lateralis was chosen to be studied for the entirety of the investigation on a subject-by-subject basis in attempt to eliminate any possible effect for limb dominance via experimental design. After each set, participants were asked to provide an RPE corresponding to the 0–10 omnibus resistance exercise scale. For clarity, all cuffs associated with the BFR exercise were modulated by a rapid inflation system consisting of two, independently controlled E20 Rapid Cuff Inflators (Hokanson, Bellevue, WA) with a shared AG101 air source (Hokanson). To provide insight into the EPR related responses, BP was collected immediately before the cuffs were inflated and deflated with a patient monitor (Datascopes Passport 2 Patient Monitor). That is, the BP of each participant was assessed before any leg cuff inflation or exercise started, and it was again assessed after the last repetition before the leg cuffs were deflated (i.e., buildup of metabolic by-products still, in theory, influenced the BP value). All leg extensions were completed using a commercially available cable leg extension device (part no. 4850; Cybex International, Owatonna, MN). Testing was performed in a temperature-controlled (21°C–22°C) and ambient light-controlled laboratory environment.

Participants

A total of 34 participants were recruited and participated in this investigation. However, because of EMG and/or NIRS malfunction(s), four participants were excluded because of missing data. Thus, 30 self-reported recreationally active, young (18–30 yr) adults (15 men, 15 women) with no cardiovascular, metabolic, neuromuscular, or renal diseases were used for subsequent analysis and all data currently presented (Table 1). All participants were instructed to continue their typical dietary, hydration, and medication habits, but were asked to abstain from exercise for 48 h before each visit. In addition, each participant was asked to not start any new nutritional supplement regimen (creatine monohydrate, sodium bicarbonate, β -alanine, etc.) during the study. Every participant was scheduled for his or her visits at approximately the same time of day (± 2 h), and completed both experimental visits within 5.2 ± 2.3 d. No attempt was made to control for the menstrual cycle, but based on the health history questionnaire, 10 women reported the use

TABLE 1. Participant characteristics (mean $\pm$ SD) examined for mean sex differences.

	Men (n = 15)	Women (n = 15)	P
Age (yr)	22 $\pm$ 3	22 $\pm$ 2	0.84
Height (cm)	177.5 $\pm$ 7.6	163.6 $\pm$ 5.3	<0.01
Weight (kg)	76.5 $\pm$ 11.8	59.5 $\pm$ 10.4	<0.01
BMI (kg·m ⁻²)	23.7 $\pm$ 3.2	22.3 $\pm$ 4.3	0.32
Lean body mass (kg)	63.4 $\pm$ 9.3	45.6 $\pm$ 5.1	<0.01
Fat mass (kg)	13.6 $\pm$ 4.7	13.5 $\pm$ 4.7	0.96
% body fat	17.6 $\pm$ 4.5	23.2 $\pm$ 4.7	<0.01
Systolic BP (mm Hg)	124.1 $\pm$ 5.6	113.8 $\pm$ 5.5	<0.01
Diastolic BP (mm Hg)	67.8 $\pm$ 5.3	66.1 $\pm$ 5.3	0.38
Bilateral 1RM (kg)	96.2 $\pm$ 11.4	61.2 $\pm$ 12.6	<0.01
Average thigh circumference (cm)	56.7 $\pm$ 4.6	52.5 $\pm$ 3.7	0.01
Average occlusion pressure (mm Hg)	149.6 $\pm$ 12.7	137.6 $\pm$ 14.2	0.02

There were no mean interlimb differences ($P > 0.05$) for circumference or occlusion pressure for the men or the women.

of pharmaceutical contraceptives. Based on previous findings (32–36) and the purpose of the current study, it was our interpretation that the likely variation in menstrual cycle phases did not substantially affect the results of the present study, especially considering the randomization procedures.

Power Analysis

Limited data relating to the interaction of sex and cuff pressure currently exist (4,6,12). We have previously observed sex differences in MAP and relevant NIRS-derived measures (e.g., oxy[heme] and deoxy[heme]) during a free-flow isometric handgrip task (25), which, based on the partial eta squared, yielded an effect size $f = 0.40$, suggesting a total sample size of 30 (power = 0.8). Freitas and colleagues (15) previously reported mean normalized EMG AMP values for men and women on a set-by-set basis, which was used to calculate an effect size of $f = 1.75$, and ultimately recommended that a smaller total sample size was necessary relative to our MAP- and NIRS-based sample size calculations. In an attempt to ensure an adequate sample size for sufficient power, the conservative approach was used, and we aimed to recruit a total of 30–40 participants as stated in the IRB application. Of note, however, our sample size calculations were largely based on free-flow isometric exercise responses, which may be limited in their ability to estimate effect sizes for dynamic, BFR exercise responses.

Familiarization Visit

The participants were given the opportunity to review and complete all documentation associated with this study (health history, informed consent, etc.). Each participant's height, weight, resting BP, and dominant leg (based on kicking preference) were recorded. Subsequently, the participants were asked to lay in the supine position on a padded treatment table and instructed to quietly rest. The research team then began the procedures for determining the minimum pressure necessary to completely occlude each femoral artery (i.e., AOP). After this, a vascular occlusion test (VOT) was conducted on the leg that was chosen, via randomization, to be used for recordings of EMG and NIRS signals. Before the initiation of the VOT, a

digital ultrasound device (Logiq E R7-Next Gen; GE Healthcare, Chicago, IL) was used to quantify adipose tissue thickness to justify the appropriateness of the NIRS device anatomical location (men = 0.53 ± 0.2 cm, women = 0.81 ± 0.3 cm). Of note, however, Ryan and colleagues (37) have stated that an ischemic calibration (i.e., VOT-based normalization) eliminates any effect of adipose tissue thickness. Once the AOP and VOT procedures were completed, the body composition of each participant was quantified by a Fit3D body scan (38) (FIT3D® ProScanner™, Redwood City, CA). Lastly, the 1RM of a bilateral, leg extension was measured following adequate warm-up and instruction. During this 1RM testing, strong verbal encouragement was given and an EMG signal was recorded from the vastus lateralis (same leg as VOT testing) for subsequent normalization procedures.

Occlusion Pressure Determination

The minimum pressures (mm Hg) needed to occlude the right and left femoral arteries were determined by systematically inflating a cuff wrapped around the upper thigh while monitoring the blood flow in the ipsilateral popliteal artery via Doppler ultrasound (Logiq E R7-Next Gen; GE Healthcare). The cuff was first inflated to 50 mm Hg to ensure proper ultrasound placement and acoustic coupling. Immediately after confirming the presence of blood flow, the cuff was deflated and then reinflated in 50-mm Hg increments until complete occlusion was identified. Once this pressure (e.g., 150 mm Hg) was identified, the cuff remained inflated and the pressure was progressively decreased until blood flow through the popliteal artery was reestablished. Next, to find the precise minimum value needed for complete occlusion, pressure was increased 1 mm Hg per 2 s until blood flow was once again not evident. If the minimum AOP of the right and left legs differed by more than 10%, this procedure was repeated.

Vascular Occlusion Test

The VOT (i.e., the ischemic calibration) conducted during the familiarization visit was used to normalize raw NIRS signals collected during the testing visits. This test was performed with Hokanson provided cuffs placed around the upper most portion of the thigh and around the ipsilateral ankle of the leg chosen to be used for NIRS and EMG measures. After a rest period, the upper and lower cuffs were simultaneously and rapidly (<0.3 s) inflated to a suprasystolic value (~250 mm Hg). The total occlusion at the level of the femoral artery was maintained for 5 min, whereas the NIRS device on the vastus lateralis recorded relative changes. At the 5-min time point, only the upper cuff was deflated to provoke postocclusive reactive hyperemia while also preventing blood from pooling in the foot vascular and thus isolating the quadricep musculature to provide accurate assessment of peak values. This is a widely accepted protocol, which has routinely been used in the related literature (39).

One-Repetition Maximum

The participants were fitted to the DCER leg extension device, and the seat was adjusted such that the lateral epicondyle of the femur aligned with the axis of rotation. The warm-up involved each individual performing 2–3 sets consisting of 3–12 submaximal repetitions at loads corresponding to 40%–80% of their anticipated 1RM. Next, based on subjective feedback, the load was progressively increased until the participant could no longer complete a leg extension through the full range of motion. The heaviest load lifted across the entire range of motion was defined as the 1RM.

Testing visits

After at least 24 h after familiarization, the participants completed an assessment of the physiological changes that occur as a result of low-load BFR leg extensions. The participants were again asked to sit on the leg extension device. Once in a rested state, BP was recorded before any BFR-related cuffs were inflated. After this and while still in the seated position, cuffs (SC12L™ 12 × 124 cm; Hokanson Inc., Bellevue, WA) were positioned around the upper most portion of each thigh. Each cuff was inflated to either 40% or 60% AOP (pending randomization) determined for each leg during the familiarization visit. Once the cuffs were positioned and inflated, the participants began the acute exercise bout. The exercise protocol matched the commonly reported BFR scheme (1,3) and included 75, low-load (30% 1RM), DCER, bilateral, leg extensions performed across four sets (1 × 30, 3 × 15; 30-s interset rest). All leg extensions were performed through the full 90° range of motion at a controlled pace (1 s concentric, 1 s eccentric). Throughout the exercise bout, NIRS and EMG were continuously collected. BP was again collected immediately after the last repetition of the last set *before* the leg cuffs were deflated. In addition, the participants provided self-reported RPE values (0–10) after each individual set, which in this context were likely influenced by the subjective intensity of effort, strain, discomfort, and/or fatigue experienced during the exercise bout (40). That is, because specific instructions were not given to dissociate exercise-induced pain/discomfort from conscious sensations of how hard, heavy, and strenuous the task was (41–44), the current measures of RPE likely were, at least in part, influenced by afferent feedback and more reflective of “exertion” as opposed to exclusively “effort” (45,46). After the exercise, the participants were monitored for 10 min to ensure adequate recovery and then scheduled for their last visit (40% or 60% AOP).

Physiological measures and outcome variables

To test our hypotheses, our methodology included noninvasively assessing NIRS- and EMG-derived parameters.

Near-Infrared Spectroscopy

During each visit, the NIRS device was worn on the surface of the vastus lateralis of either the dominant or nondominant limb (16 participants used dominant, 14 participants used

nondominant). The device used in the current study was a portable, dual-wavelength (760 and 850) continuous-wave, spatially resolved NIRS instrument (Portamon; Artinis Medical Systems, Elst, the Netherlands). Based on previously reported successful approaches (23,47), it was attached at 75% of the distance from the anterior superior iliac spine to the lateral portion of the patella using black self-adherent wrap (3M, Coban). It should be noted that the Portamon only provides arbitrary changes in light attenuation (relative, not absolute values (in grams per deciliter)). Using this device, oxy[heme] and deoxy[heme] were quantified (39). These NIRS variables were determined in accordance to the modified Beer–Lambert Law under the assumption that the differential path length factor (DPF) and G (loss of light due scattering) remained constant. The DPF factor was set to 4 for all participants. In addition, the data were collected at the maximum sampling rate of the device (10 Hz) and reported from the Tx2 channel (optode distance of 35 mm). While using this device, the signal quality was continuously assessed by viewing the percentage of light reaching the photodiode and the signal fit factor. To examine sex differences (between factor) in the NIRS-derived outcome variables, values were reported relative to the ischemic calibration (i.e., VOT) performed during the familiarization visit. That is, all values were presented as a percentage of the respective range. We have previously reported adequate reliability of the range collected during the VOT used for this normalization (48). The normalization equation (25,48) used was:

$$\% \text{ of physiological range} = \frac{[\text{value} - \text{minimum}]}{\text{range}} \times 100 \quad [1]$$

Surface Electromyography

For muscle excitation (18), a wireless sensor (27 × 37 × 13 mm, Delsys Avanti™; Delsys, Inc., Natick, MA) was placed on the same vastus lateralis as the NIRS device. The center of this sensor was attached with transparent surgical tape (Med PRIDE Transparent Medical Tape) at 66% of the distance between the anterior superior iliac spine and the lateral superior border of the patella (i.e., just proximal to NIRS). The sensor was oriented such that the longitudinal axis was parallel to the angle of pennation of the vastus lateralis (20°). Before the placement of the sensor, the skin was dry shaved and cleaned with isopropyl alcohol, which in theory reduced interelectrode impedance and increased the signal-to-noise ratio. The contact material of the sensor is reported to be 99.9% silver, consist of an inter-electrode distance of 10 mm, and have a mass of 14 g. The common mode rejection ratio was >80 dB, and the sampling rate was set to 1926 samples per second within the TrignoLite acquisition unit (Delsys, Inc.). All signals were collected during the 1RM testing and bilateral BFR leg extension bouts, and specifically, during the concentric phase of each muscle action, which was visibly tracked in real time with Delsys and later aligned with our offline signal processing. Specifically, the concentric portions of each muscle action were processed offline (Labview version 2021; National Instruments, Austin, TX) using a fourth-order Butterworth bandpass

(10–499 Hz) filter. The EMG AMP (root mean square, in microvolts) values were calculated across the concentric phase of each muscle action (epoch approximately 1 s (1926 data points)) and normalized to the 1RM values (i.e., expressed as percent of 1RM). In addition, the 1RM values were also assessed across the same 1-s epoch.

Statistical Analysis

Aligned with previous research (9,15,49), difference scores were calculated to assess the physiological changes that occurred during each set. Specifically, the average of the first three repetitions and the average of the last three repetitions were calculated, and subsequently, the change was determined (average of last three – average of first three). This eliminated the time factor nested within the set factor, which would have prompted the need of four-way ANOVA models. Therefore, separate three-way, sex × set × cuff pressure mixed factorial ANOVA models were used to examine mean differences in EMG AMP, oxy[heme], and deoxy[heme]. In addition, separate mixed-factorial ANOVA models were used to assess potential mean differences in MAP (sex × time × cuff pressure) and RPE (sex × set × cuff pressure). Of note, the ANOVA for MAP and RPE assessed the absolute values, not change scores. Significant interactions were decomposed with either follow-up paired/independent *t*-tests or additional ANOVA. In the absence of a significant three- or two-way interaction, main effects were evaluated with Bonferroni-corrected pairwise comparisons. For all Bonferroni-corrected comparisons, the corresponding α value is presented. Ninety-five percent confidence intervals (CI_{95%}) of the mean difference were presented alongside all significant findings. Partial eta squared (η_p^2) was used as measures of effect for each ANOVA model. The Greenhaus–Geisser correction was applied when sphericity was not met (assessed with Mauchly's test of sphericity). All calculations and statistical analyses were conducted utilizing Statistical Package for the Social Sciences software (version 26.0; IBM Inc., Chicago, IL) and GraphPad Prism (version 8.3.4; GraphPad Software, San Diego, CA). A *P* value ≤ 0.05 was considered statistically significant. All data were reported as mean ± SD.

RESULTS

EMG AMP responses. There was no significant three-way or any two-way interactions for EMG AMP, but there was a significant main effect for set ($P < 0.001$, $\eta_p^2 = 0.273$). Collapsed across sex and cuff pressure, the pairwise comparisons indicated that the change in EMG AMP exhibited during set 1 (14.8% ± 5.6%) was significantly ($P = 0.001$ –0.004) less than set 2 (22.6% ± 8.6%; $\Delta\text{CI}_{95\%}$, 2.0%–13.5%), set 3 (27.0% ± 11.9%; $\Delta\text{CI}_{95\%}$, 4.1%–20.2%), and set 4 (26.0% ± 11.8%; $\Delta\text{CI}_{95\%}$, 3.4%–18.9%) as depicted in Figure 1. There were no other ($P > 0.05$) differences between sets. In addition, there was not a significant ($P = 0.077$, $\eta_p^2 = 0.107$) main effect for sex or cuff pressure ($P = 0.666$, $\eta_p^2 = 0.007$).

NIRS responses. For oxy[heme], there was no significant ($P > 0.05$) three-way or any two-way interaction, but there

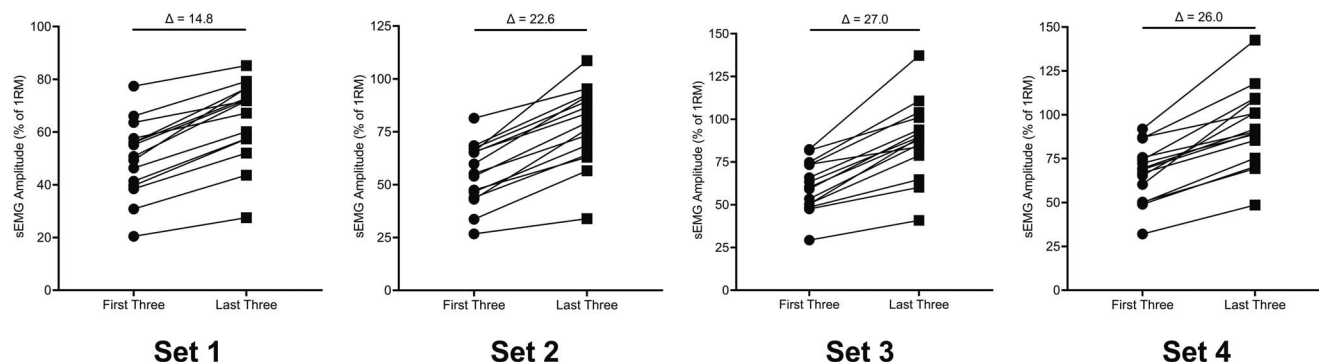


FIGURE 1—The plots display the individual changes ($n = 15$) collapsed across sex and cuff pressure from the beginning to the end of each set. The Δ reports the mean difference. There was a significant ($P < 0.05$) effect for set such that the change in EMG AMP during set 1 was significantly less than all other sets.

were main effects for cuff pressure ($P = 0.032$; $\eta_p^2 = 0.154$) and set ($P < 0.001$; $\eta_p^2 = 0.540$). The pairwise comparisons indicated that (collapsed across sex and set) there was a greater change for 40% AOP compared with 60% condition ($-24.6\% \pm 19.7\%$ vs $-20.7\% \pm 18.4\%$; $\Delta CI_{95\%}$, 0.37% – 7.5%). In addition, (collapsed across sex and cuff pressure) as shown in Figure 2, the greatest change in oxy[heme] was exhibited during set 1 ($-40.1\% \pm 22.6\%$), whereas the smallest decrease was demonstrated during set 2 ($-12.2\% \pm 7.6\%$).

For deoxy[heme], there was not a significant three-way interaction ($P = 0.678$; $\eta_p^2 = 0.013$), but there was a significant ($P = 0.002$; $\eta_p^2 = 0.210$) cuff pressure–set interaction (collapsed across sex). Based on the purpose of the study, we chose to decompose the interaction with four, paired t -tests to examine mean differences between the two cuff pressures at each Set. As seen in Figure 3, during sets 2–4, there was a significantly greater change ($\sim 6\%$) for 40% AOP condition than 60% AOP. There was no significant ($P = 0.153$) difference between the conditions at set 1 ($46.6\% \pm 16.7\%$ vs $51.3\% \pm 18.3\%$, respectively).

BP responses. For MAP, there was no significant three-way or any two-way interaction, but there were significant main effects for time (preexercise to postexercise; $P < 0.001$, $\eta_p^2 = 0.852$) and sex ($P = 0.001$, $\eta_p^2 = 0.312$). There was, however, no significant main effect for cuff pressure ($P = 0.150$, $\eta_p^2 = 0.073$). Collapsed across sex and cuff pressure (Fig. 4A), the pairwise comparisons indicated that there was a significant ($P < 0.001$; $\Delta CI_{95\%}$, 14.8 – 19.3 mm Hg) increase in MAP from preexercise (87.5 ± 4.3 mm Hg) to postexercise (104.5 ± 4.1 mm Hg). In general, (collapsed across time and cuff pressure), the men (99.5 ± 5.5 mm Hg) exhibited a significantly greater ($P < 0.001$; $\Delta CI_{95\%}$, 2.6 – 11.4 mm Hg) MAP than women (92.5 ± 5.3 mm Hg). Furthermore, only collapsed across cuff pressure, the men increased MAP by $21.1\% \pm 10.3\%$, whereas the women exhibited an $18.3\% \pm 7.7\%$ increase.

Ratings of perceived exertion. The three-way ANOVA indicated that there were no significant interactions or a main effect for sex. However, there were significant main effects for cuff pressure ($P = 0.042$, $\eta_p^2 = 0.140$) and set ($P < 0.001$, $\eta_p^2 = 0.842$). The pairwise comparisons indicated that 60% AOP provoked a significantly greater RPE response (collapsed across sex and set) than 40% AOP (8.3 ± 0.5 vs 8.0 ± 0.6 , respectively; $\Delta CI_{95\%}$,

0.013 – 0.670). In general, (collapsed across sex and cuff pressure), there was a significant ($P < 0.001$) increase across set (Table 2).

DISCUSSION

The main objective of the present study was to evaluate the physiological responses resulting from a commonly utilized acute, BFR, resistance exercise bout with two different cuff pressures in healthy men and women. Our main findings included the following: 1) regardless of cuff pressure or sex, the change in muscle excitation from the first to the last repetition increased within each set; 2) in general, the 40% AOP cuff pressure elicited greater intraset changes in oxy[heme], whereas there was also a progressive decrease in the magnitude of change in oxy[heme] across sets; 3) excluding the first set, the 40% AOP cuff pressure demonstrated the greatest intraset changes in deoxy[heme]; 4) regardless of cuff pressure, there were uniform increases in MAP, but the women generally presented with lower MAP; and 5) the 60% AOP condition provoked greater RPE responses, and as expected, there was an increase in RPE across sets regardless of AOP.

In contrast to our hypothesis and a similar study of exercise-induced changes in EMG AMP, the present results suggested that sex did not influence the magnitude of increase in muscle excitation from the beginning to the end of each set. Previously, Freitas and colleagues (15) suggested that there may be differences in EMG AMP between men and women during low-load BFR, but unlike the current investigation, their analysis only evaluated a single value of EMG AMP (e.g., global average) during a particular set, not the change across that set. Although their work is meaningful in numerous aspects such as by providing evidence of a possible EMG AMP-related sex difference and testable hypotheses, it offered limited temporal resolution, especially considering here we showed that, on average, the EMG AMP values increased $\sim 23\%$ during each set. Furthermore, in terms of cuff pressure, our results complemented the previous findings of Singer et al. (17), who reported that 11 men did not exhibit different degrees of increase in EMG AMP during a single set of 30 unilateral, isokinetic, leg extensions with cuff pressures corresponding to 60%, 80%, and 100% AOP. It should be highlighted,

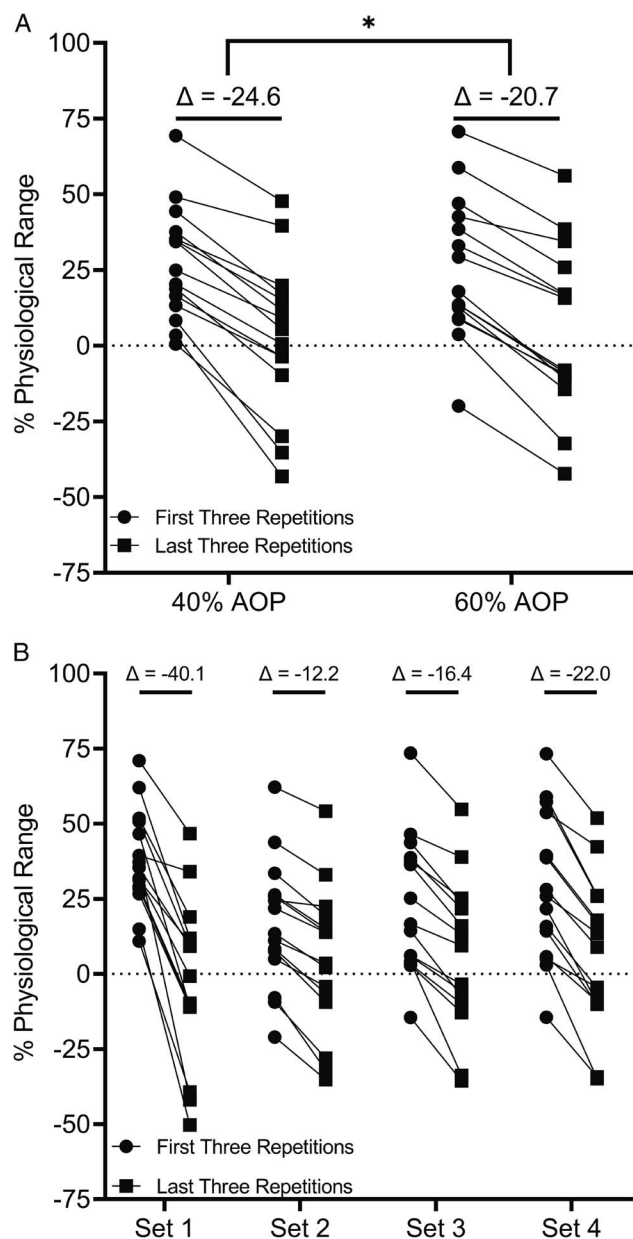


FIGURE 2—A, The plots display the individual changes ($n = 15$) collapsed across sex and set from the beginning to the end of each set. The Δ reports the mean difference. There was a significant ($*, P < 0.05$) difference in the mean change scores such that the 40% AOP condition demonstrated a greater decrease in oxy[heme] than the 60% AOP condition. **B,** Collapsed across sex and cuff pressure, the greatest decline from the beginning of the set to the end of the set in oxy[heme] was exhibited during set 1. The set configuration consisted of 30–15–15–15 repetitions with 30 s of interset rest.

however, that this previous investigation (17) differed from the current study in many ways including the inclusion of women, number of sets, and exercise modality (bilateral vs unilateral and DCER vs isokinetic). In general, our collective EMG AMP results suggested that across the four sets of BFR leg extensions, the magnitude of intraset changes in muscle excitation increased after set 1. Of note, there seemed to be a similar increase in EMG AMP during sets 2–4, perhaps suggesting that, under these conditions of BFR, a maximum adjustment was reached after two sets

(Fig. 1). To be clear, the absolute load used for the leg extensions was not altered during the exercise bout, so the observed increases in muscle excitation were not a function of increased motor unit recruitment in response to a new, heavier load. Thus, during each set, the increase in muscle excitation could have been driven by a combination of some other factors like increased efferent drive, reflective of greater effort/exertion (50), to combat various symptoms of fatigue by increasing motor unit recruitment, firing rate, and/or motor unit synchronization, all of which are known to influence values of EMG AMP (1,23,51). Although not directly tested in the current investigation, it is also possible that the “oxygen conforming” phenomenon contributed to our change in EMG AMP because of the possible attenuation of oxygen delivery and subsequent increase in muscle excitability compared with a hypothetical free-flow condition (22), which has been observed and reported by Yasuda et al. (20,21). In addition to the presumable reduction in oxygen delivery, during BFR resistance exercise, there is a diminished ability to clear metabolic by-products, and this subsequent accumulation results in the stimulation of group III/IV muscle free nerve endings, likely propagating greater afferent feedback (3,19,20,52,53). As a proof of concept, we recently reported that BFR (60% AOP) combined with 75 repetitions at 30% 1RM resulted in greater impairments distal to the neuromuscular junction (i.e., peripheral fatigue due to accumulation of by-products) than central factors when compared with a matched non-BFR exercise condition (1). Taken together, we hypothesize that some degree of excitation coupling was impaired via by-products such as inorganic phosphate (19), but because of the blockage of venous outward flow, the by-products continued to accumulate triggering greater increases in muscle excitation (20) during subsequent sets until a maximum increase was reached. It is still curious that there were no sex-specific EMG AMP responses considering that men and women may exhibit morphological differences that could affect fatigue processes including substrate utilization, metabolite buildup, and muscle excitability (14,23,54,55). Alternatively, our findings corroborated the notion that cuff pressure does not result in EMG AMP differences as long as venous return is uniformly occluded during largely anaerobic tasks like resistance exercise (9,17,20,49,56). We do, however, acknowledge that future work remains needed to confirm our findings with other approaches such as motor unit decomposition software.

Beyond muscle excitation, the oxy[heme] and deoxy[heme] results provided further insight into the underlying physiological consequences that occur during acute BFR resistance exercise with two different cuff pressures. Like EMG AMP, the current results failed to support our stated hypothesis that men and women would exhibit unique NIRS-derived responses. This was surprising based on our previous reports (25,48) that have demonstrated men desaturate at a faster rate in rested and fatigued states as well as during low-load isometric exercise. Specifically, here we showed that regardless of sex or set, the 40% AOP condition yielded the greatest overall intraset changes in oxy[heme]. Initially, this was perceived as

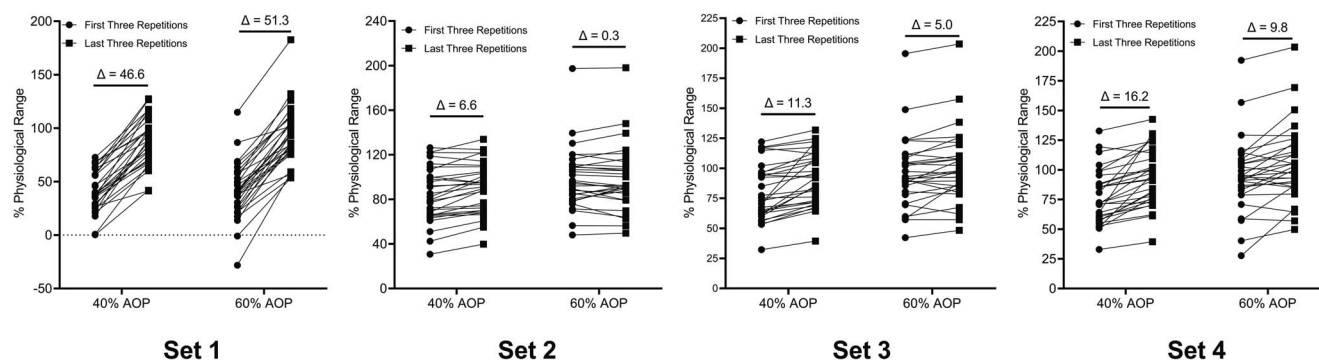


FIGURE 3—The plots display the individual changes (collapsed across sex; $n = 15$) from the beginning to the end of each set. The Δ reports the mean difference. The exercise protocol included four sets with a repetition configuration defined as 30–15–15–15. For deoxy[heme], there was a significant difference between the conditions indicating that there were greater increases during the 40% AOP vs 60% AOP trials, except for set 1, which resulted in no significant difference between the cuff pressures.

counterintuitive; however, it was our eventual interpretation that this was suggestive of a reduced ability to recover throughout the interset rest during the 60% AOP compared with 40% AOP condition. In other words, during the interset rest of the 40% AOP condition, participants were at least partially able to resaturate (i.e., recover in terms of oxy[heme]), but in comparison, during the 60% AOP condition, this ability to recover was attenuated. This is depicted in Figure 2A such that despite a greater overall change for the 40% AOP condition (24.6% vs 20.7%), the mean values corresponding to first and last three repetitions of the 60% AOP condition were lower. In contrast, as a result of only analyzing a single set of 30 repetitions, Singer et al. (17) showed that cuff pressures ranging from 0% to 80% AOP did not affect oxy[heme] responses. Perhaps, under those conditions, the actual exercise (not the manipulation of blood flow) was a potent enough stimulus to decrease oxy[heme], which ultimately masked any measurable differences related to the application of a cuff. This was evidenced by their 0% AOP condition (17), which resulted in similar oxy[heme] responses as compared with the trials with various >0 mm Hg cuff pressures. Thus, a likely key characteristic underlying the unique benefits of BFR lies within the interset rest. For example, the current oxy[heme] findings supported the rationale of the “30–15–15–15 protocol” because of the considerable decline during the first set, which was never replicated during subsequent sets (Fig. 2B). This is again indicative of an inability to fully recover during the 30 s of interset rest, regardless of cuff pressure, creating at least a quasi-hypoxic environment to hypothetically drive positive, exercise-induced adaptations (57,58). An exciting future direction includes the continued evaluation of positive adaptations resulting from BFR applied only during rest phases.

The current deoxy[heme] responses complemented the oxy[heme] findings such that the 40% AOP and 60% AOP cuff pressures elicited different magnitudes of change in deoxy[heme] across the BFR exercise bout. However, there was a similar magnitude of the increase in deoxy[heme] between the two cuff pressures as a result of set 1. Said differently, the initial 30 repetitions (regardless of cuff pressure and sex) generated considerable metabolic stress (25,59), which, in

theory, had to be tolerated during the remaining three sets. As the participants continued, however, it was again our interpretation that less recovery occurred during the interset rest of the 60% AOP condition compared with the 40% AOP. This can be seen in Figure 3 such that during sets 2–4, the 40% AOP trial elicited a greater increase in deoxy[heme] than the 60% AOP. It should be noted, however, that it seemed that the 60% AOP condition remained at an elevated level of deoxy[heme] even after each rest period as demarcated by the values of the first three repetitions for sets 2–4 (i.e., starting sets at nearly 100% of the calculated physiological range). Previously, Biazon et al. (59) reported a relationship between deoxy[heme] and muscle cross-sectional area after low-load BFR training, which they understood as support for the theory that metabolic stress promotes hypertrophy. Based on Figure 3, it seemed that the values corresponding to the last three repetitions of each set were similar, suggesting that the 40% AOP and 60% AOP conditions likely ended the individual sets at approximately the same level of deoxy[heme], despite possible different starting points (i.e., first three repetitions). Similarly, Suga et al. (60) demonstrated with 12 participants (6 men) that there was no differences in metabolic stress (e.g., intramuscular metabolites and pH) as a result of BFR exercise with various cuff pressures at the same relative low load. Thus, the question remains whether it is necessary to maintain the high levels of deoxy[heme] (e.g., proxy for metabolic stress) after set 1 or if the slight recovery afforded by lower cuff pressures may still provide comparable adaptations (4). Most recently, Das and Patton (4) stated that cuff pressures less than 50% AOP may cause “...insufficient metabolic stress...” but the current results may suggest otherwise. Reis et al. (7) also suggested that cuff pressures greater than 40% AOP should be used, but their results indicated that deoxy[heme] did not change across the BFR exercise for any condition except for the 40% AOP and non-BFR trials. In support of our findings, previous BFR training studies have used 40% AOP to elicit adaptations such as 1) increased muscle size and strength after 12 wk of BFR resistance training at 40% 1RM + 40% AOP (8), 2) increased muscle size and strength after 8 wk of unilateral forearm flexion at 30% 1RM + 40% AOP, and 3) increased isometric and isokinetic strength as well as muscle

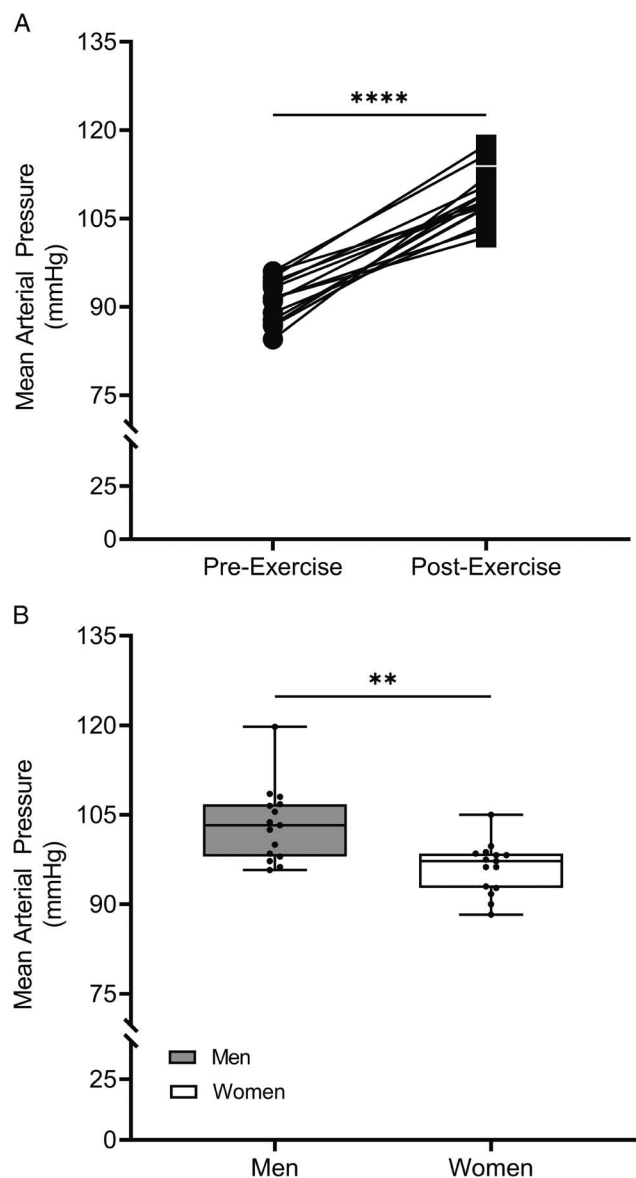


FIGURE 4—A, The plots display individual changes ($n = 15$) in MAP values (collapsed across sex and cuff pressure) from preexercise to postexercise. The Δ reports the mean difference. There was a general significant ($P < 0.05$) increase in MAP from preexercise to postexercise. **B,** The box-and-whisker plots display MAP values (collapsed across cuff pressure and time) for the men and the women. The “whiskers” extend to the minimum and maximum points, and the horizontal line demarcated the median. The results indicated that the men exhibited a significantly ($P < 0.05$) greater average MAP value.

thickness in response to 8 wk of unilateral leg extension at 15% 1RM + 40% AOP. Of note, in most cases, the 40% AOP cuff pressure did not significantly differ from higher cuff pressure difference in terms of magnitude of benefit, except perhaps there may be a greater improvement in muscular endurance when using a higher cuff pressure (8–10). Therefore, these inconsistencies in cuff pressure recommendations only add to the growing rationale for future work to determine if there are meaningful differences in chronic, BFR training-induced adaptations between groups using cuff pressures of 40% AOP or higher with equal relative, low-loads.

Metabolic stress is a known contributor to the EPR such that the exercise-induced accumulation of metabolic by-products activates group III/IV muscle afferent free nerve endings to increase cardiovascular parameters such as MAP in part to maintain adequate perfusion to vital organs. Here, we reported that, regardless of sex and cuff pressure, there was ~19% increase in MAP after the BFR exercise bout. The timing of these measurements likely influences the overall observed magnitude of change, and thus, it should again be considered that our BP values were collected before the cuff inflation as well as before the deflation of the cuffs after the completion of the exercise bout. The current NIRS-derived responses supported this general increase in MAP considering that there was no difference between the men and women for deoxy[heme]. It was again curious that there was not a statistically significant sex difference in the percent increase in MAP (men vs women, 21% vs 18%) considering the strong previous evidence (26,35), including from our research group (25), indicating that women typically present with an attenuated EPR. Most of the previous related work, however, did not utilize a BFR exercise model, and therefore, perhaps the interpretations resulting from isometric muscle actions such as handgrip do not extend to BFR conditions. Moreover, it is possible that, like EMG AMP, a maximum stimulus was achieved given the lack of ability to clear metabolic by-products, which yielded a non-sex-specific increase in MAP. It should also be noted that there was not a cuff pressure-specific increase in MAP, suggesting that the 40% AOP and 60% AOP conditions provoked a similar *metaboreflex* considering that there was not a difference in the load (i.e., *mechanoreflex*) between trials (19,25,26). This adds to the speculation that if venous return is uniformly occluded and arterial flow is not completely compromised, there will likely be similar acute physiological responses such as increases in MAP, especially if the measurement is assessed before the release (i.e., deflation) of the cuff. In terms of this MAP increase, Cristina-Oliveira et al. (24) presented concerns regarding the potentially augmented EPR during low-load BFR. Within this review (24), however, it was highlighted that when low-load BFR exercises are compared with high-load, free-flow conditions, the data are not clear such that only 50% of previous studies have shown a greater EPR response (i.e., greater BP) after low-load BFR. Based on their report and the current findings, we acknowledge that low-load BFR likely does not circumvent *all* of the known potential concerns related to high-load resistance exercise, particularly in cardiac rehabilitation patients, but there have been promising advancements (5). Overall, BFR exercise is intended to prevent the clearance of metabolic by-products during a given bout, which, under the current conditions, led to no-sex or cuff pressure-specific changes in MAP, ultimately suggesting that healthy men and women can expect similar MAP responses when utilizing cuff pressures ranging from 40% to 60% AOP.

TABLE 2. RPE across set (collapsed across sex and cuff pressure).

$n = 30$	Set 1	Set 2	Set 3	Set 4
RPE (0–10)	6.7 $\pm$ 0.7	7.9 $\pm$ 0.6	8.8 $\pm$ 0.4	9.3 $\pm$ 0.5

As stated previously, in terms of expected exercise-induced adaptations, it still remains to be fully elucidated whether or not it is necessary to maintain a constant high level of metabolic perturbations with higher cuff pressures (e.g., 60% AOP) compared with permitting slight recovery during rest periods by using lower pressure like 40% AOP. Intuitively, it is most advisable to use the lowest effective cuff pressure during BFR exercise training in nearly all settings (athletic, clinical, etc.) to mitigate potential negative side effects while simultaneously maximizing enjoyment (2,24,29). There is likely an exercise-induced discomfort and corresponding degree exertion interaction with future physical activity adherence, which is of particular relevance because of higher cuff pressures generally resulting in higher self-reported values of RPE and/or discomfort (27–29,61). The current results indicated that collapsed across set and sex, the 60% AOP condition exhibited ~4% greater RPE responses than the 40% AOP cuff pressure. In addition, regardless of sex and cuff pressure, there was a linear increase in degree of perceived heaviness and strain experienced during the BFR bout as demarcated by the RPE values (40,42). It should again be mentioned that deoxy[heme] values were similar between the cuff pressures at the end of each set, which was when the RPE values were collected. Under the current conditions, as outlined in the Methods section, we did not provide specific instructions to the participants to dissociate pain/discomfort from perceptions of effort (43), and therefore, the current RPE findings were likely at least partially influenced by afferent feedback from group III/IV free nerve endings stimulated in the active skeletal muscle (44,46). Thus, it is possible that the current RPE responses were more in line with perceived *exertion* as opposed to exclusively perceptions of *effort* (41,42,45,50), although there has been a previous critique related to the distinction between effort and exertion (62). Taken together, our results, in conjunction with previous reports, suggest that the lower cuff pressure most likely yields lower RPE values in healthy men and women, albeit perhaps only slightly lower, because of possible differences in magnitude and consistency of metabolic stress–related afferent feedback. This may prove to be an important consideration when attempting to maximize enjoyment and subsequent adherence to exercise (29).

When evaluating our discussed interpretations, readers should consider the following aspects of our experiment. First, we did not use neuromuscular techniques capable of identifying specific locations of impairment (e.g., peripheral vs central fatigue) or normalize the EMG AMP signal to M-waves, which may offer additional value particularly related to inferences of efferent drive (18). Also, we did not continuously assess MAP or collect these values after each individual set, which reduces our temporal resolution. This investigation also analyzed sex as a biological factor, which should not be considered a strength but rather a necessity in our view. We did not, however, control for the menstrual cycle or use of pharmaceutical contraception, which may be interpreted as a limitation. Although we respect and acknowledge this, variations in the menstrual cycle have been shown to have minimal influence on

baroreflex sensitivity, autonomic modulation, or NIRS-derived values (33). In addition, most previous investigations do not support a role for the menstrual cycle in influencing the EPR (32,35). Furthermore, in terms of the appropriateness of our sex-based comparisons, here the NIRS values were normalized to an ischemic calibration (25,39,48), which is commonly ignored or substituted for other normalization methodology. Lastly, in terms of our statistical analysis approach, difference scores were used to evaluate the magnitude of change during a particular set, but this does not directly test for differences between cuff pressure or men and women specifically at baseline (e.g., initial) and/or post (e.g., final), but does provide a magnitude of change relative to each sex and permits fewer comparisons (and greater statistical power) within the ANOVA-based models (i.e., one collective value evaluated per set as opposed to two values (first three, last three)).

CONCLUSIONS

In summary, it was our main objective to present findings describing the physiological responses that acutely occurred during low-load BFR resistance exercise in men and women as a result of two different cuff pressures. With the current methodology, it was shown that sex and cuff pressure did not meaningfully influence changes in EMG AMP suggesting that a lower pressure may be acceptable if the goal is to maximize muscle excitation. There were also no sex-related differences in NIRS-derived outcomes, which indicated that men and women likely experience similar challenges along the oxygen cascade, at least under the current conditions. The NIRS parameters may have, however, provided insight that the 40% AOP cuff pressure allowed for partial recovery during the interset rest, but regardless, still likely elicited similar metabolic stress (as demarcated by deoxy[heme] values [25,59]) by the end of each set. Furthermore, here we did not report that cuff pressures uniquely invoked the EPR, which may be encouraging to some clinicians and practitioners concerned with unfavorable exercise-induced increases in BP, but in general, the 40% AOP was perceived to be easier in terms of RPE. Overall, the current study found no evidence that a cuff pressure greater than 40% AOP should be used, except perhaps if the maintenance of metabolic stress is necessary, as well as that healthy men and women will likely respond similarly to matching BFR exercise prescriptions.

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